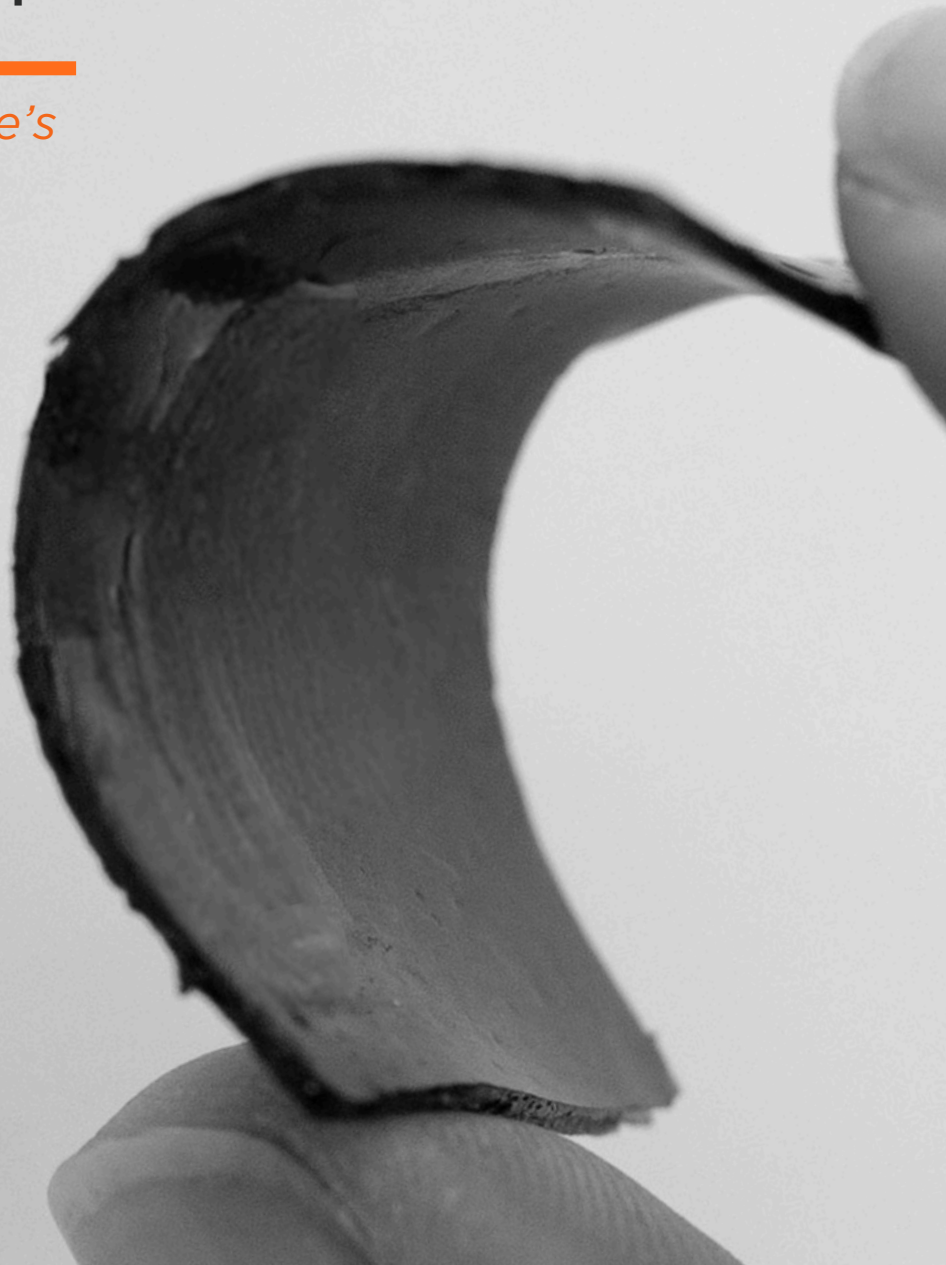


Hy-AEGIS Nanocomposites

**Graphene based composites with high performance
for EM shielding Apps**

*Unlocking Graphene's
Full Potential*



Coating Systems Based on N-Graphene Hybrids for Electromagnetic Shielding via Absorption

Context:

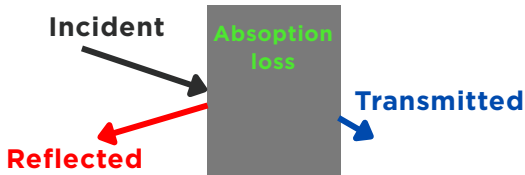
The electromagnetic (EM) pollution emerged as a global issue, as a result, of the rapid growth of electronic and communication devices. The widespread popularization of portable devices, 5G or future 6G wireless communication technologies inevitably generate a substantial amount of EM radiation with different frequencies.

The EM pollution provokes problems such as system failures, data loss, signal disturbance, and causes a serious threat to communication security and most importantly human health.

The Hy_AEGIS systems utilize nitrogen-doped graphene (N-graphene) hybrid materials embedded into polymer matrix to achieve effective electromagnetic shielding primarily through absorption mechanisms (> 60%) rather than reflection in a broad 5–70 GHz frequency range encompassing the C, X, Ku, K, Ka, Q and V microwave bands.

Traditional metallic shielding reflects electromagnetic waves, which can create secondary interference. In contrast, N-graphene hybrid coatings are designed to absorb and dissipate electromagnetic energy as heat, reducing reflected radiation and improving electromagnetic compatibility. The system can be applied to flexible, rigid substrates with different geometries using conventional brushing techniques.

Are all EM shielding materials equal?

			
Reflection	High		Low
Transmission	Low		Low
Absorption loss	High		High
Reflection loss	High		Low
Total EMI shielding efficiency	High	} Reflection - dominated	High
EM absorption	Low		High
Stealth	✗		✓

Why it works?

Dielectric loss

Conduction, dipole polarization, interfacial polarization

Magnetic loss

Natural resonance, exchange resonance, eddy current loss

Optimized and tailored filler architecture

Maximizing absorption, minimizing secondary EM pollution

Market place:

Advanced electromagnetic N-graphene-based Hy_AEGIS functional nanocomposites can be utilized in wearable electronics, Wi-Fi systems, satellite communications, and radar stealth technologies.

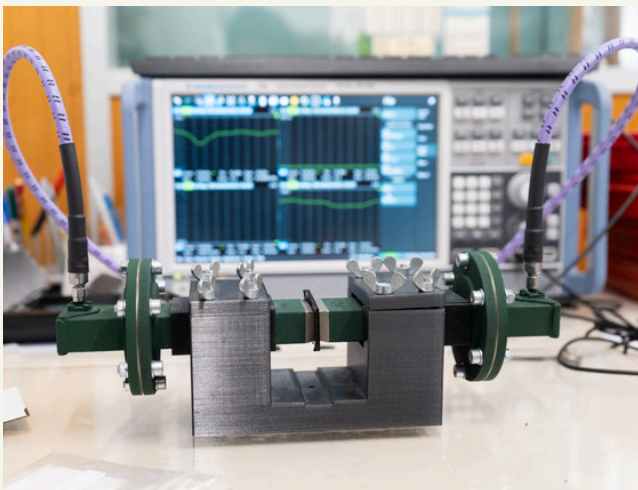
These electromagnetic functional materials provide enhanced shielding performance primarily through absorption mechanisms, effectively protecting devices and systems from external electromagnetic interference while improving operational reliability and signal stability.

Utilization:

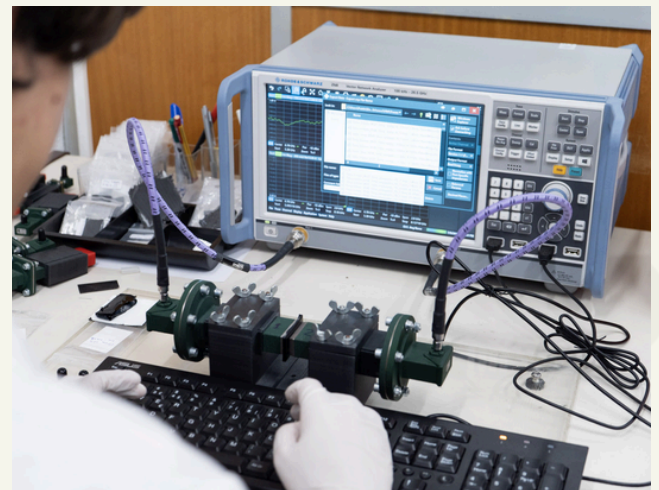
The coating can be applied through a simple two-step process:

1. Mixing the Hy_AEGIS powder uniformly with a polymer system.
2. Applying the resulting composite coating onto the target surface using a paint brush.

After air drying, the coating forms a durable electromagnetic shielding layer with absorption-dominated EM protection properties.



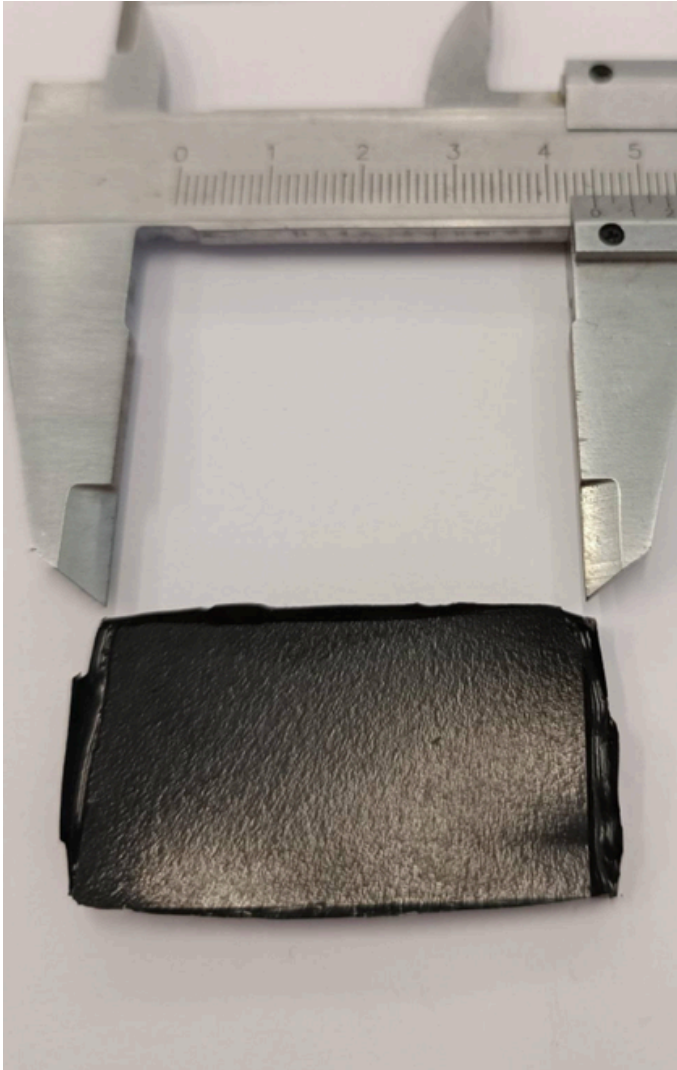
Material absorption characterization



Hy_AEGIS Nanocomposites for High-Performance Electromagnetic Shielding

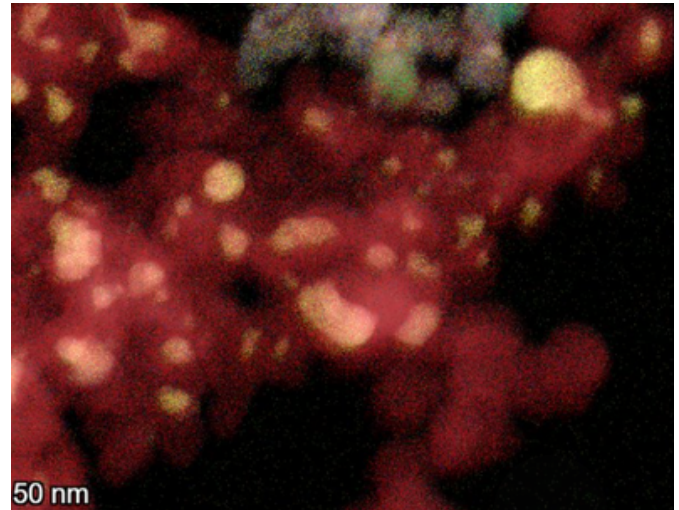
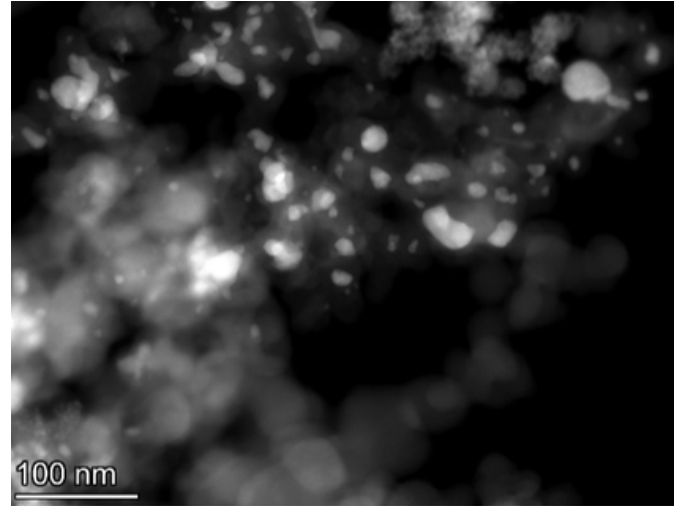
Technical Info

Nanocomposite Hy_AEGIS on conducting substrate:



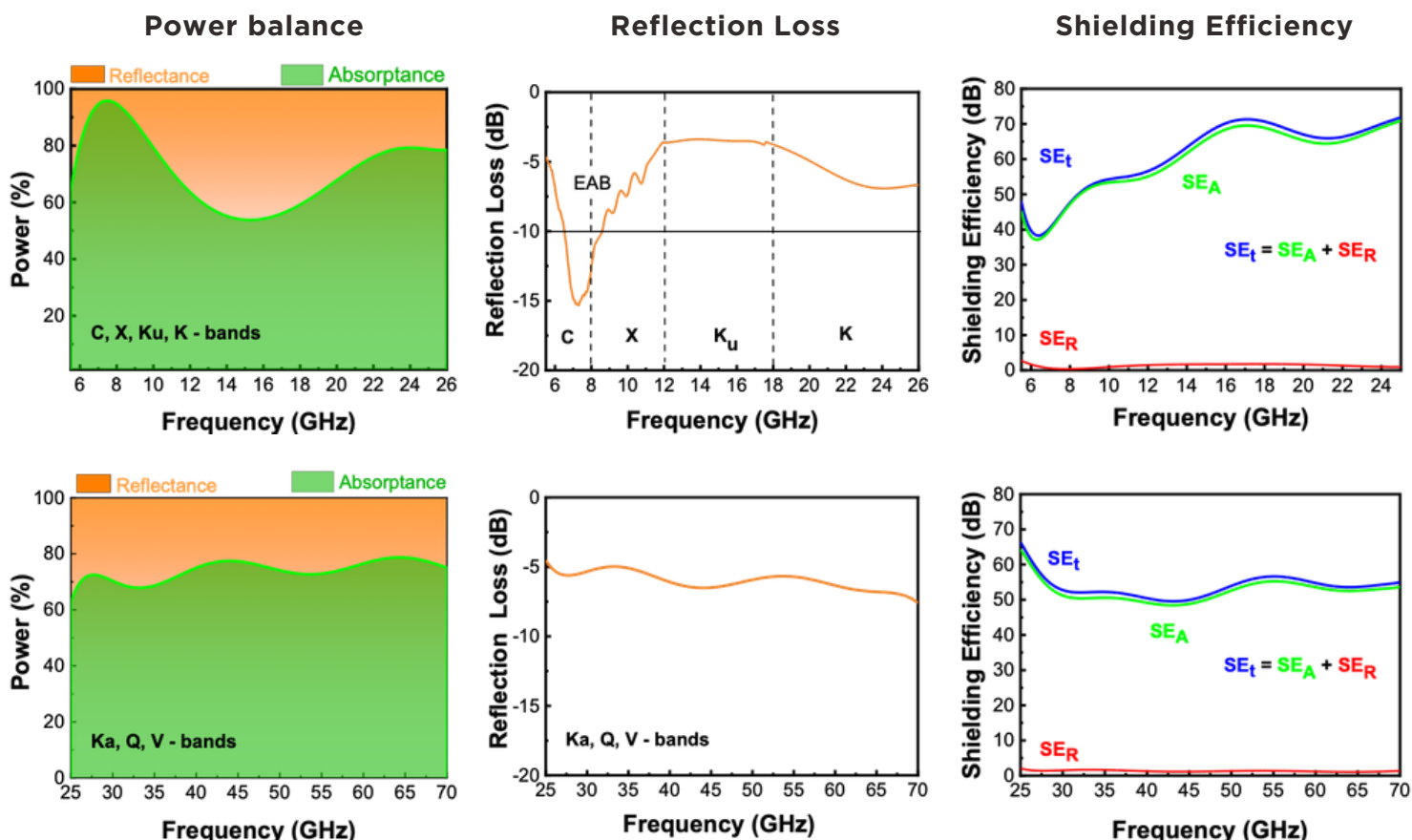
Deposition thickness 2 mm

STEM images of N-graphene based hybrids:



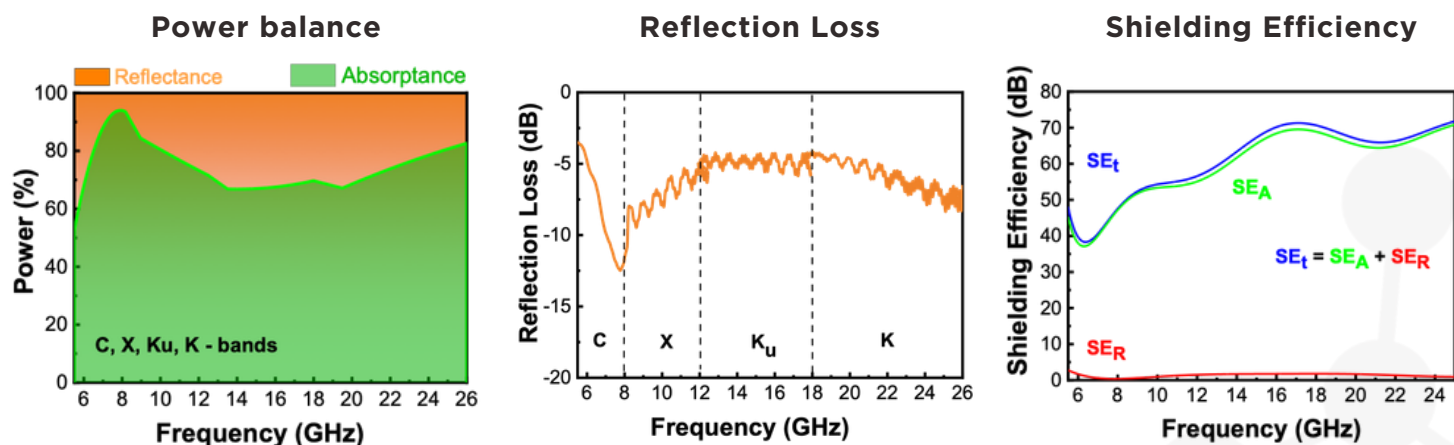
Electromagnetic Shielding Performance of the Hy-AEGIS-01 Nanocomposite

All the measurements were taken in 2 mm sample graphene hybrid on Epoxy (15 wt. %) over conductive substrate.



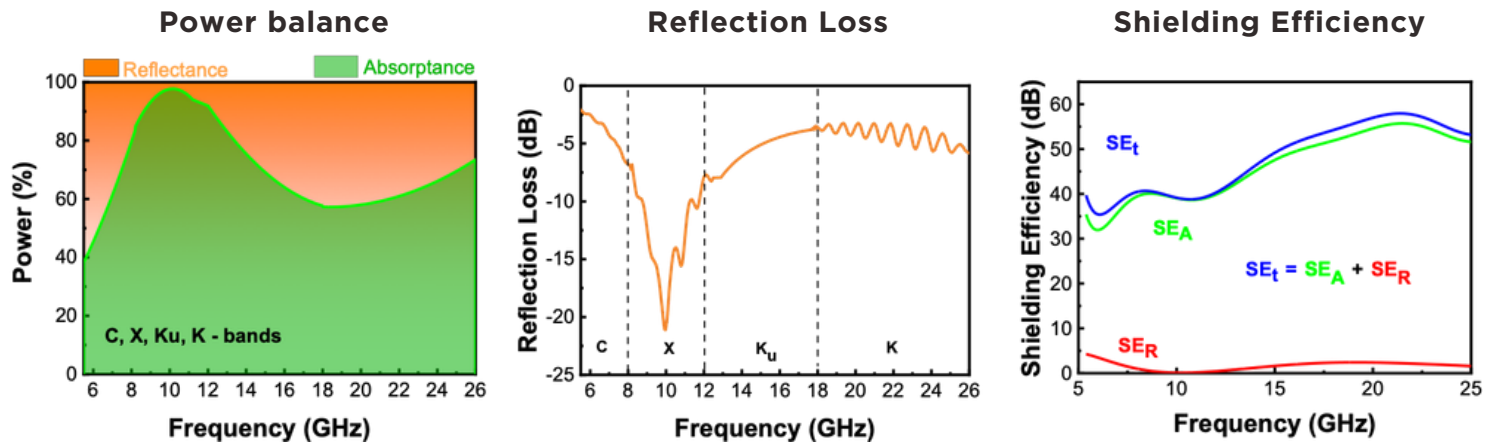
Achieves absorption efficiencies above 60% in the C, X, Ku, and K frequency bands (5-26 GHz).
Delivers absorption efficiencies exceeding 70% in the Ka, Q and V frequency bands (25-70 GHz).

Electromagnetic Shielding Performance of the Hy-AEGIS-02 Nanocomposite



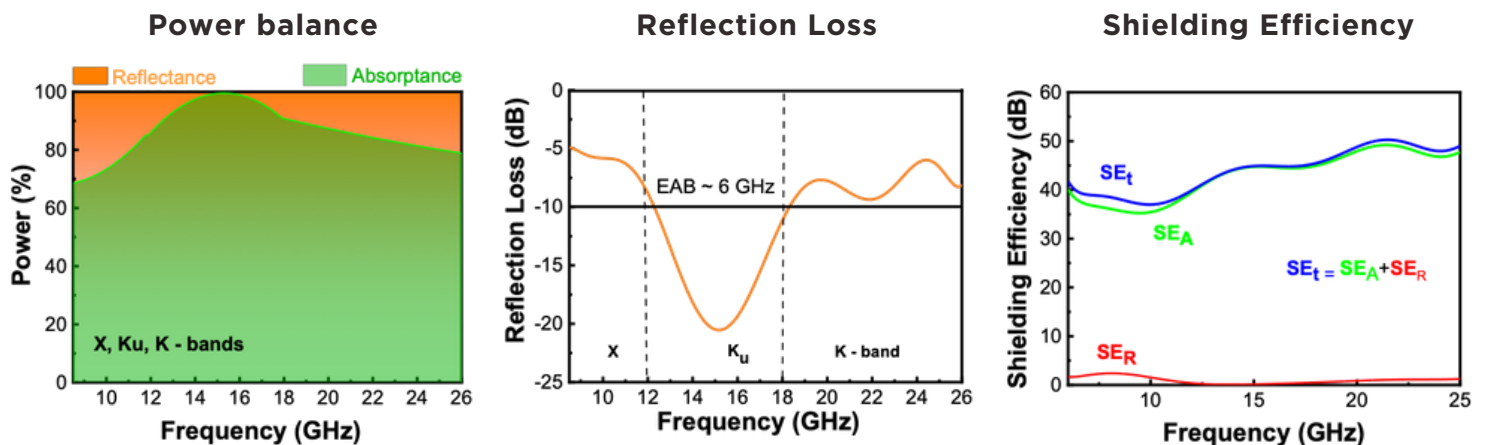
Absorption efficiency exceeds 70% across the C, X, Ku, and K bands (5-26 GHz).

Electromagnetic Shielding Performance of the Hy-AEGIS-03 Nanocomposite



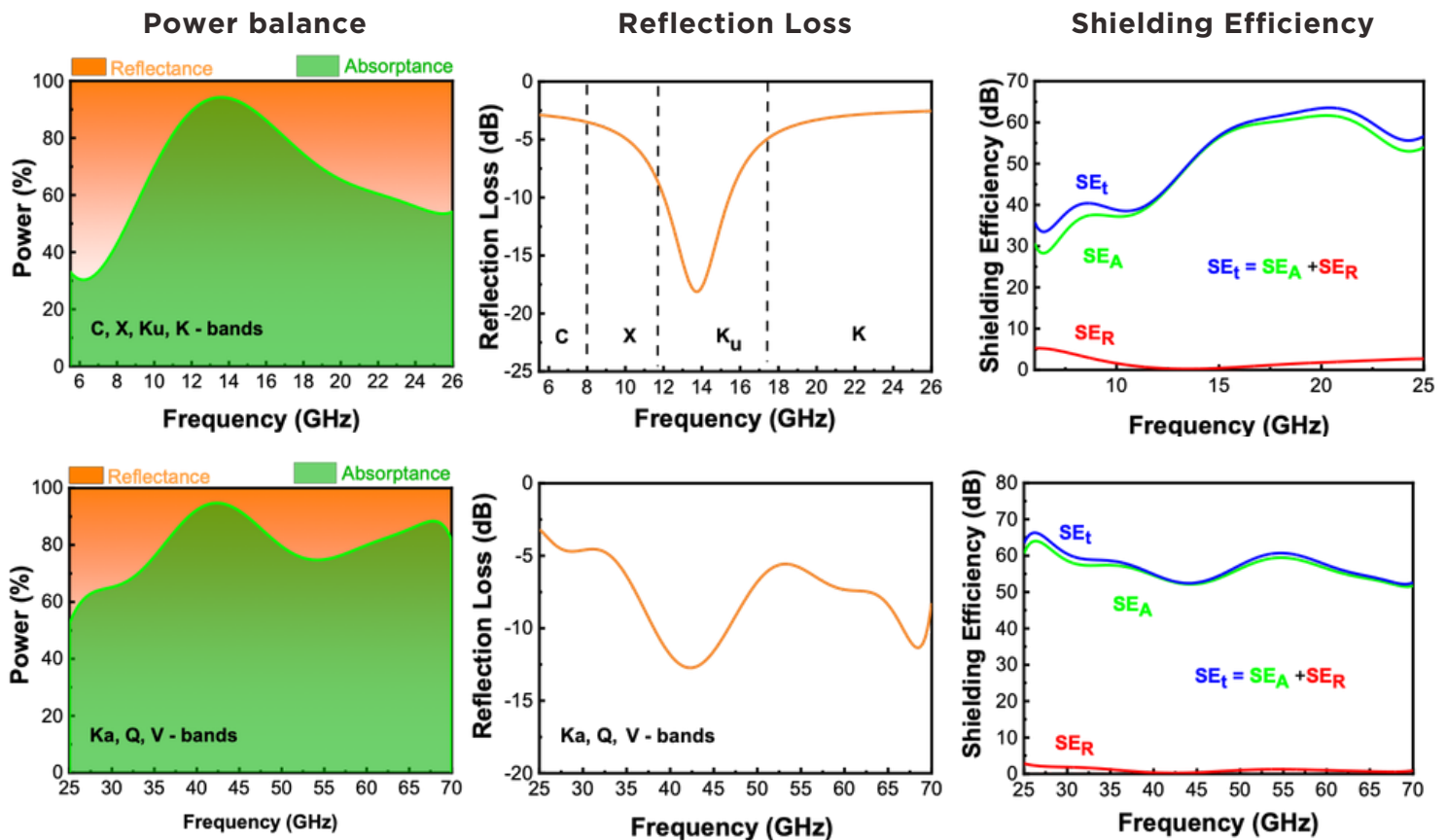
Achieves absorption efficiencies above 90% in the X band (8-12 GHz) and over 60% in the Ku and K bands (12-26 GHz).

Electromagnetic Shielding Performance of the Hy-AEGIS-04 Nanocomposite



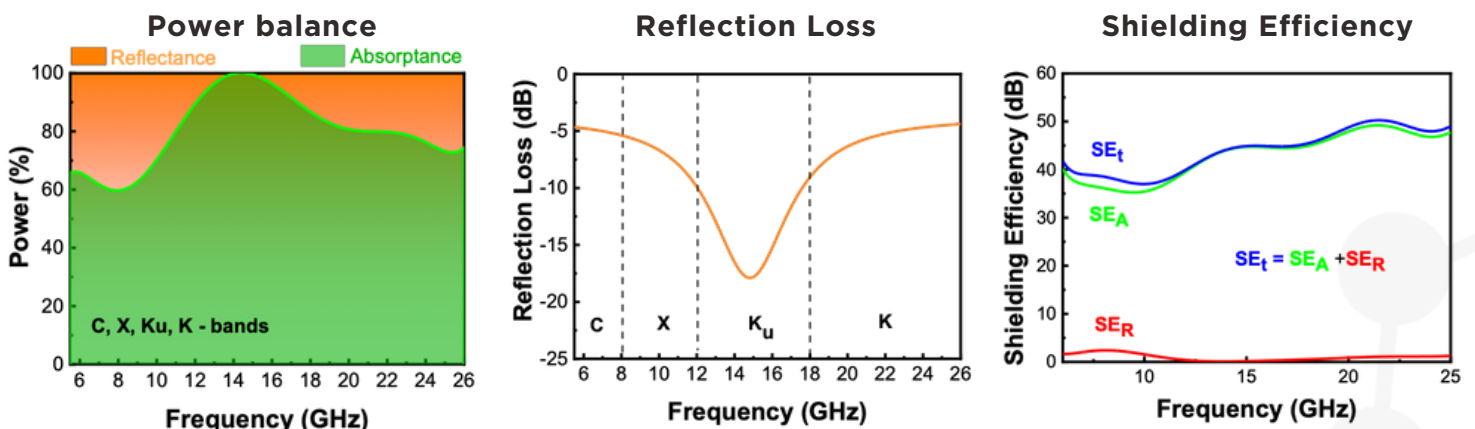
Achieves absorption efficiencies above 90% in the Ku band (12-18 GHz) and over 70% across the X (8-12 GHz) and K (18-26 GHz) bands.

Electromagnetic Shielding Performance of the Hy-AEGIS-05 Nanocomposite

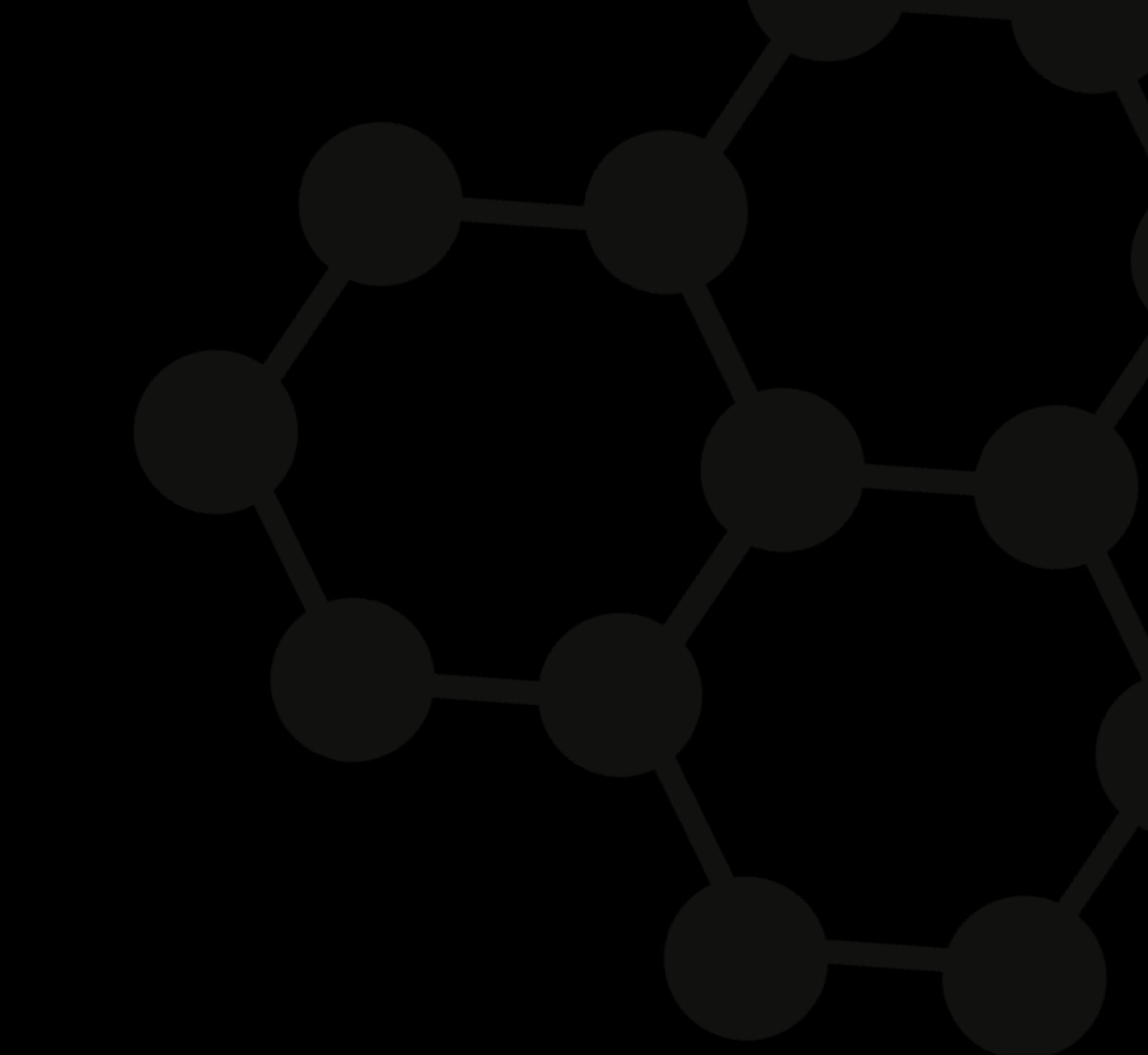


Achieves absorption efficiencies above 60% in the X, Ku, and K frequency bands (8-26 GHz). Delivers absorption efficiencies exceeding 70% in the Q and V frequency bands (35-70 GHz).

Electromagnetic Shielding Performance of the Hy-AEGIS-06 Nanocomposite



Achieves absorption efficiencies above 90% in the Ku band (12-18 GHz), 65% in the C band (5-8 GHz), and over 70% across the X band (8-12 GHz) and K band (18-26 GHz).



Unlocking Graphene's Full Potential

▾ About GTechPlasma

We produce customized high-quality graphene-based materials with enhanced properties using an innovative, versatile, cost-effective and environmentally friendly plasma technology.

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